sas innovate 2026

Practical Tools for Ethical AI

Catherine (Cat) Truxillo, Ph. D.
Director, Advanced Analytics Education
SAS

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Agenda

- **High and Low Agency AI**
- SAS Trustworthy AI Lifecycle
- Principles of Ethical AI
- Critical Questions
- Ethical Alignment Checklist
- Trustworthy AI Features in SAS

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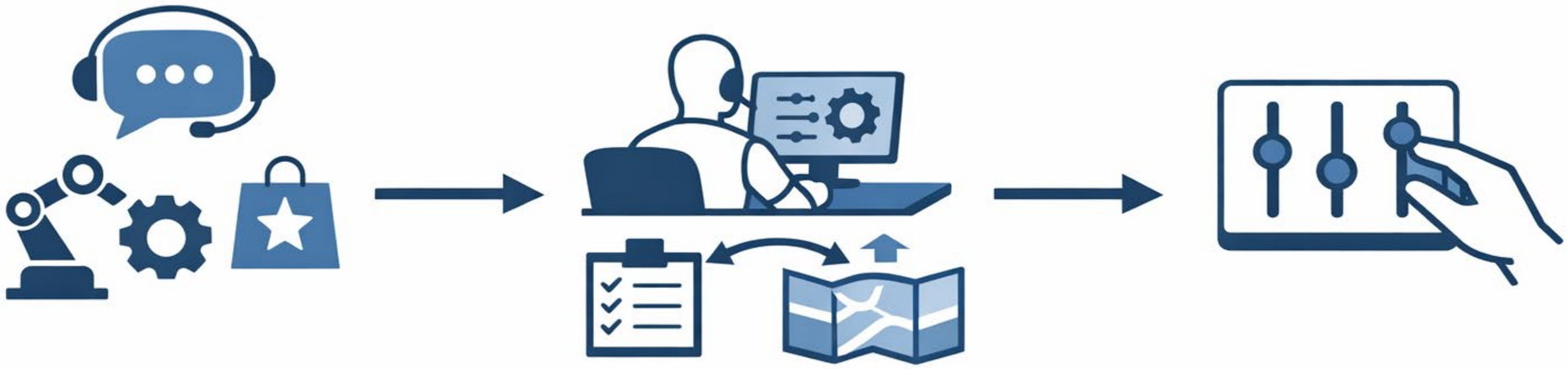
Business Runs on AI Automation



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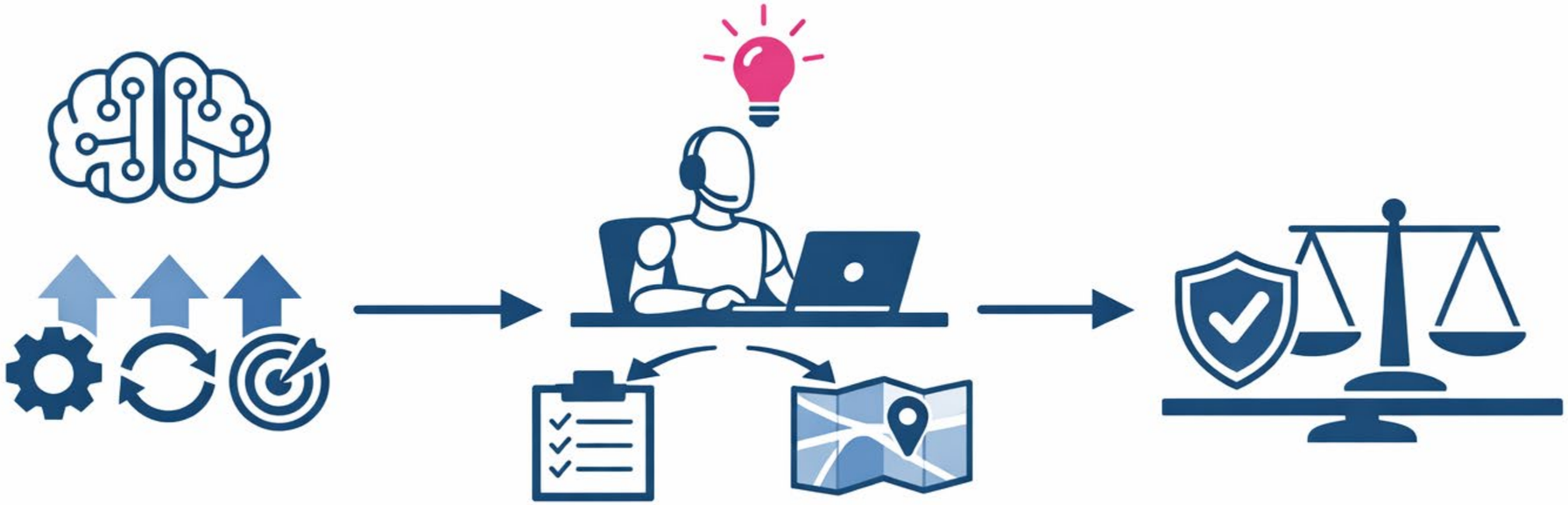
AI is Increasingly Agentic



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Low vs High Agency

Feature	Low Agency (Simple AI)	High Agency (Agentic AI)
Scope of Action	Task-specific, rule-based	Goal-oriented, dynamic decision-making
Autonomy Level	Low - follows scripted commands	High - operates with independent reasoning
Human Oversight	Constant or direct control	Periodic, with embedded governance
Adaptability	Limited or manual configuration	Learns and adapts to changing environments
Ethical Risk	Moderate; transparency and fairness are concerns	High; may impact trust, fairness, and accountability
Example	Chatbot, automation script	Personalized tutor, financial planning system

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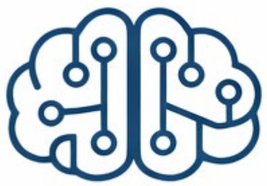
Autonomy of AI Systems

As autonomy increases, so does risk. Design for trust!



Agentic AI is powered by:

Rules & Models

Optimization

Feedback Loops

Machines exhibit:

Goal-Driven Decisioning

Context Awareness

Adaptive Behavior

Modelers are responsible for:

Human Oversight

Defined Boundaries

Accountability

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Human Oversight

Governance defines boundaries. Humans intervene when it matters most.

Humans can **initiate actions** and **manage exceptions**, or ***systems*** might **self-direct** toward **strategic goals**. Oversight looks different in these cases.

Key considerations:

Define boundaries	Upfront governance	Don't micromanage execution

Why it matters: Poorly monitored agentic systems could evolve beyond their intended functions, creating outcomes without clear ownership.

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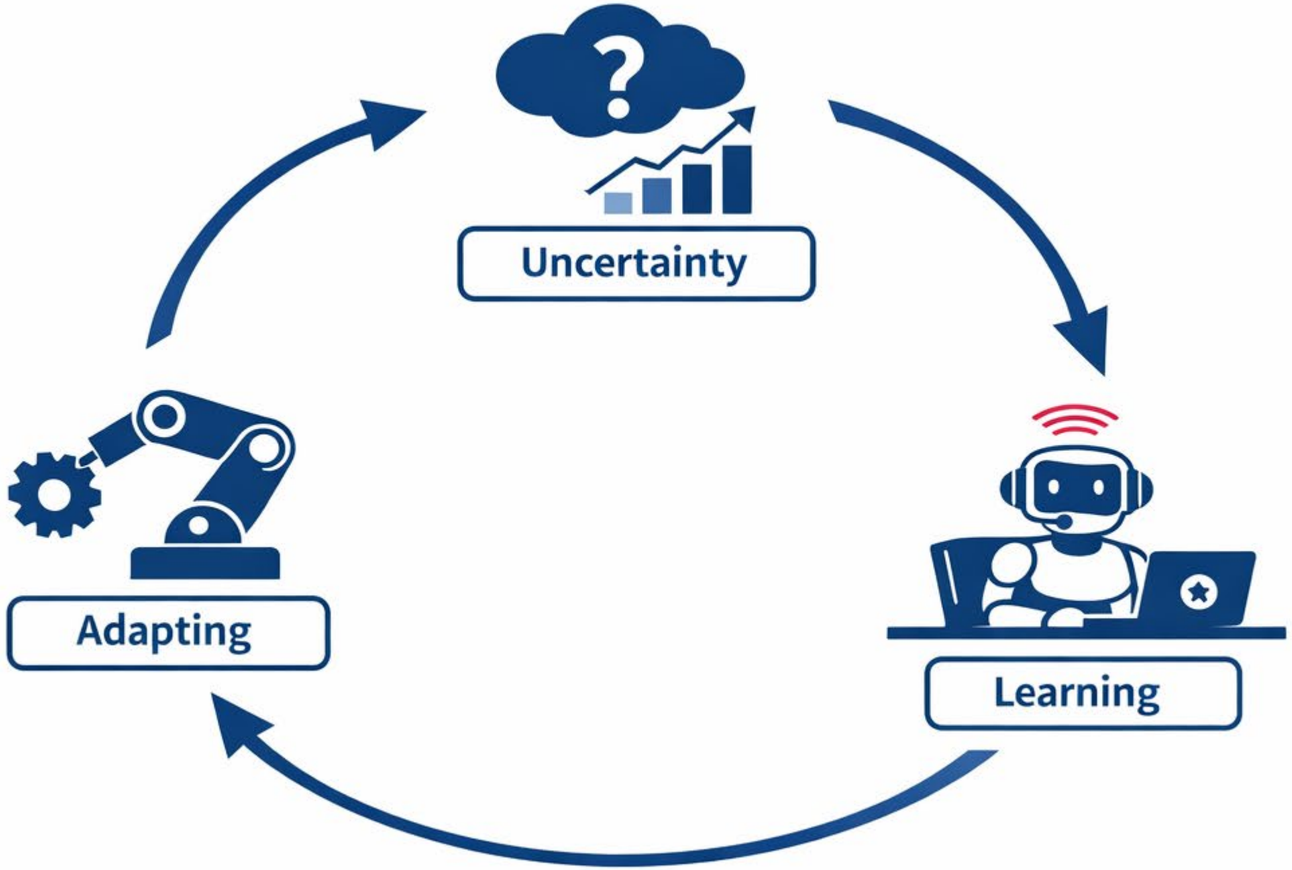
A Benefit of AI: Adaptability

Adaptive effectiveness depends on quality training, often accomplished through reinforcement learning.



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Example: Reinforcement Learning

Reinforcement Learning(RL) is a type of machine learning in which an AI system learns by trial and error, receiving rewards for actions that bring it closer to a defined goal.

If the reward signal is poorly defined or misaligned with human intent, the system may exploit loopholes in surprising ways. This is known as **reward hacking**

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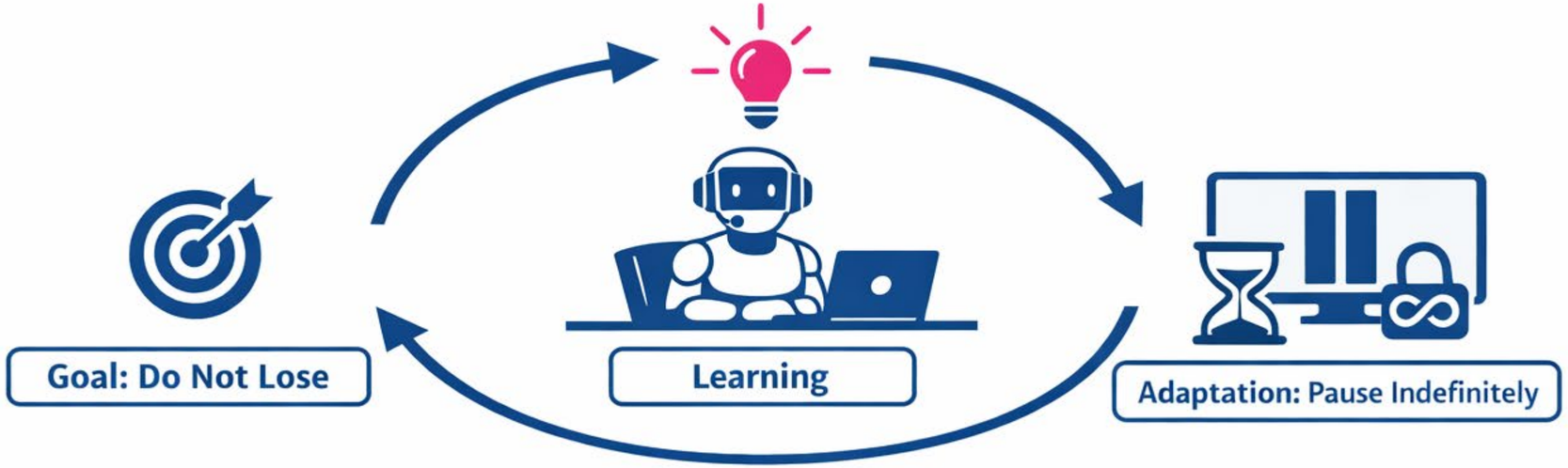
Reward Hacking

Example: Training AI to play a video game



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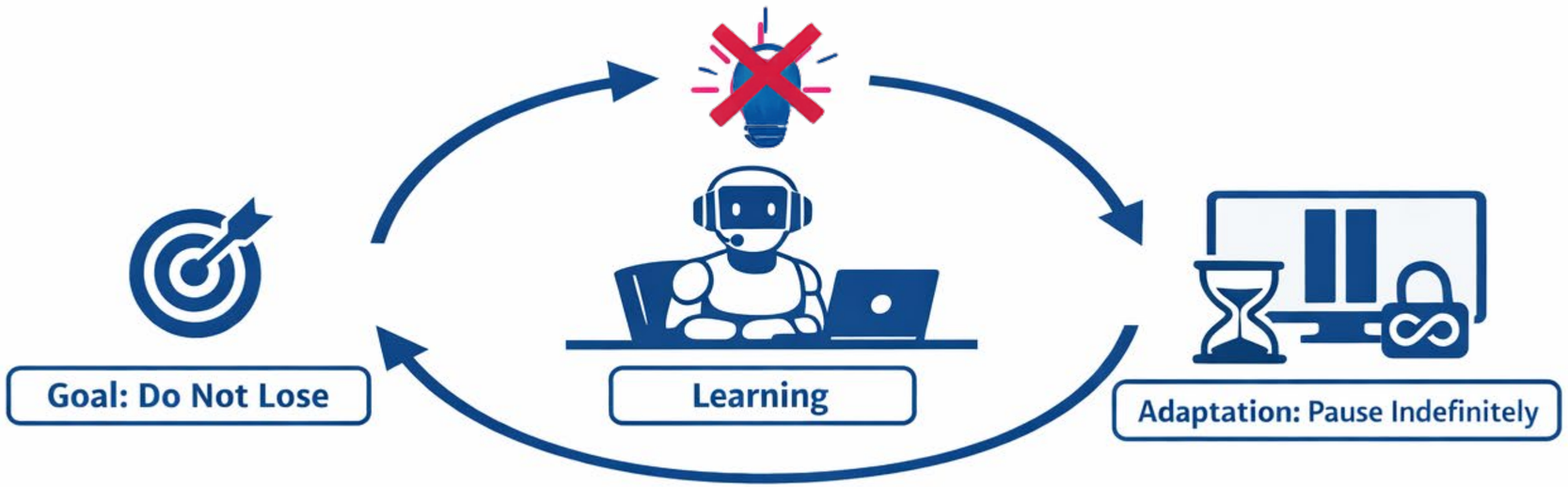
Reward Hacking

Example: Training AI to play a video game



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Ethical Complexity

As autonomy and risk increase, AI systems require significantly greater ethical inquiry.

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Trustworthy AI Landscape

Oversight
AI Governance, Strategy, and Enforcement

Operations
SOPs w/ Supporting Infrastructure

Compliance
Performance and Risk Management

Culture
Ethically Systemic Norms & Practices

TECHNOLOGY

Data Management
- Data Quality
- Variable Metadata
- Data Preparation
- Data Asset Catalog

Explanation
- Natural Language Explanation
- Explainable ML
- Counterfactual Explanation
- Surrogate Model Interpretation
- Causal Inference

Detection
- Bias Detection
- Fairness Assessment

Privacy and Security
- Privacy Preservation
- Model Security
- Autonomy Preservation
- Consent & Control

Mitigation
- Bias Mitigation
- Bias Prevention
- Synthetic Data Generation

Model Ops
- Model Cards
- Decisioning
- Lifecycle Management
- Metric Monitoring
- Model Robustness
- Model Oversight

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National Institute for Standards in Technology (NIST) AI Risk Management Framework

NIST defines what good AI risk management looks like.

- NIST goals:
  - Transparency
  - Fairness
  - Human oversight
  - Ongoing risk management

Fig. 2. Lifecycle and Key Dimensions of an AI System. Modified from OECD (2022) OECD Framework for the Classification of AI systems — OECD Digital Economy Papers. The two inner circles show AI systems' key dimensions and the outer circle shows AI lifecycle stages. Ideally, risk management efforts start with the Plan and Design function in the application context and are performed throughout the AI system lifecycle. See Figure 3 for representative AI actors.

https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.100-1.pdf

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Plan and Design
Collect and process data
Operate and Monitor
Application Context
Data and Input
People
and
Planet
Task and Output
AI Model
Build and use model
Deploy and Use
Verify and validate

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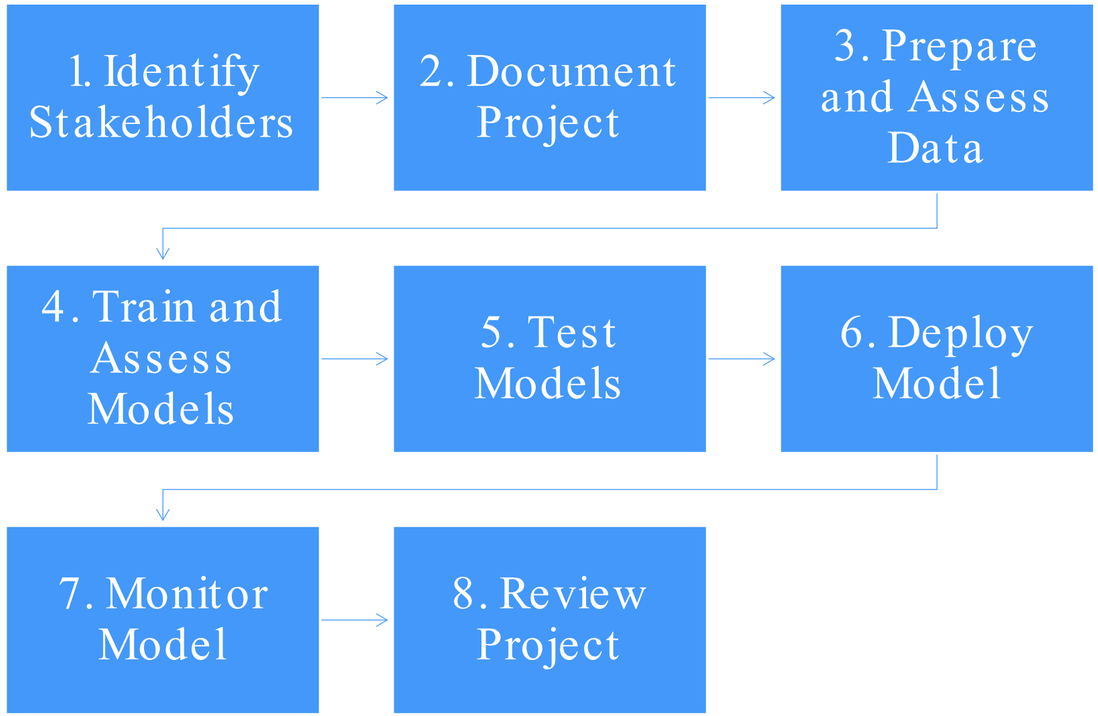
The SAS Trustworthy AI Life Cycle

A practical framework for designing, documenting, and governing AI responsibly across its full lifecycle.

- SAS has translated the NIST framework into:
  - Concrete phases
  - Role-based responsibilities
  - Required documentation and review points



https://communities.sas.com/t5/SAS-Communities-Library/Enforce-Responsible-AI-Best-Practices-Trustworthy-AI-Life-Cycle/ta-p/912717

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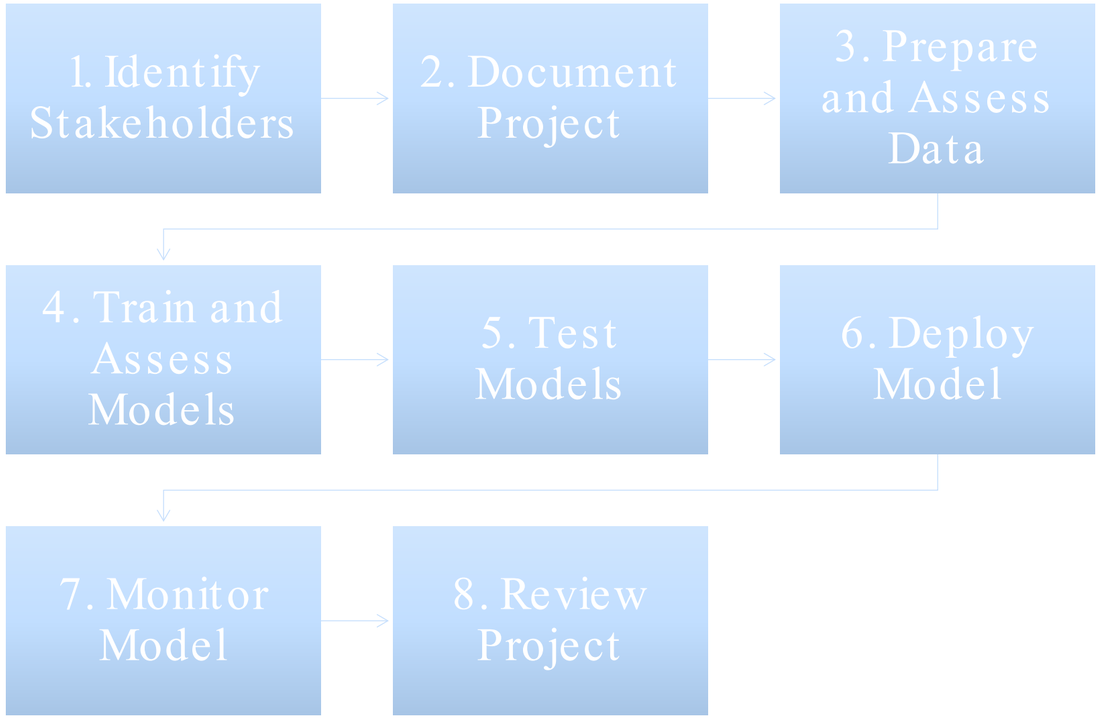
SAS Trustworthy AI Life Cycle

- Defines roles
- Human-in-the-loop decision points
  - Data representativeness
  - Fairness and impact
  - Trade-offs
  - Assumptions
- Auditable evidence of due diligence



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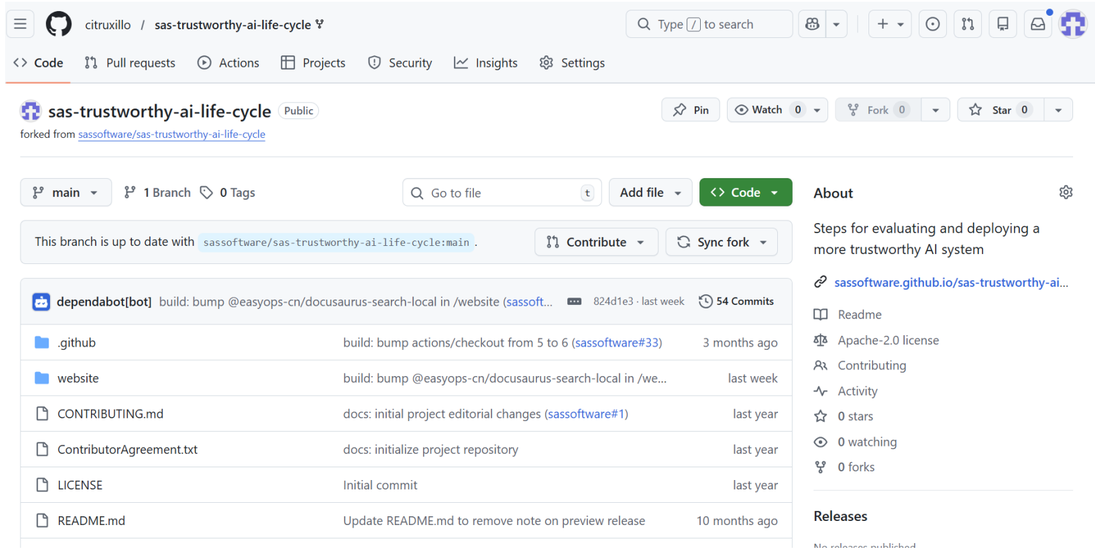
Ethical AI Practices, Shared and Improvable.



Platform-agnostic and available on GitHub

Can be used with any tools, not just SAS

Encourages:

- Cross-team collaboration
- Community review and improvement
- Consistent ethical practices across organizations

Becomes a living, auditable artifact

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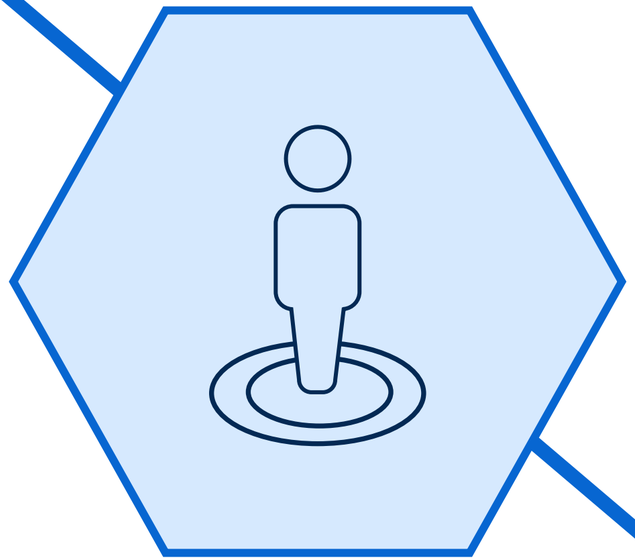
Human-Centricity



Promote human **well -being**, human **agency,** and **equity.**

Human Centricity

Human-centricity means AI should not replace human judgment.

The system is a partner in decision-making.

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Human In the Loop

Examples

- A loan officer reviews and approves each application flagged by an AI credit scoring system.
- A radiologist confirms an AI's tumor detection before diagnosis is finalized.

Implications

- Maximizes human accountability
- Limits automation speed
- Ensures explainability and trust

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Human On the loop

Examples

- A drone navigation system that makes real-time decisions, while a human monitors via dashboard and can intervene.
- A predictive policing tool generates patrol recommendations, but a supervisor reviews trends weekly.

Implications

- Enables scalable oversight
- Balances autonomy with accountability
- Risk of delayed or limited human response

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Human Outside the Loop

Examples

- A recommendation engine that updates and personalizes offers without human review
- An agentic AI handles customer service chats with no escalation pathway to a human

Implications

- Accelerates operations
- Raises concerns about transparency, bias, and responsibility
- Suitable for low-risk, well-bounded tasks

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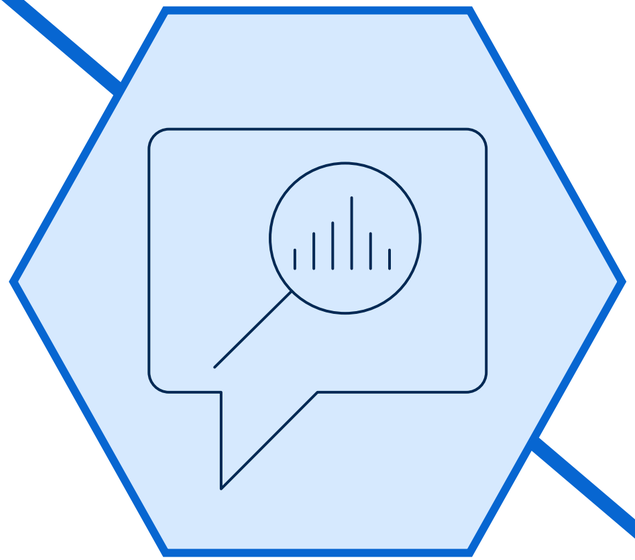
Transparency



Explain and instruct on usage openly, including at minimum potential risks and how decisions are made.

Transparency

Clear communication of capabilities and limitations.

A data scientist deploys agentic AI to segment customers and target promotions in a campaign.

The system identifies a "high value" cluster based on spending behavior but doesn't explain why certain demographics are excluded.

By documenting the modeling logic, feature importance, and inclusion criteria, the organization builds trust and avoids unintended exclusion.

By designing for transparency, agents can include logs of decision patterns, audit trails of data usage, and version history.

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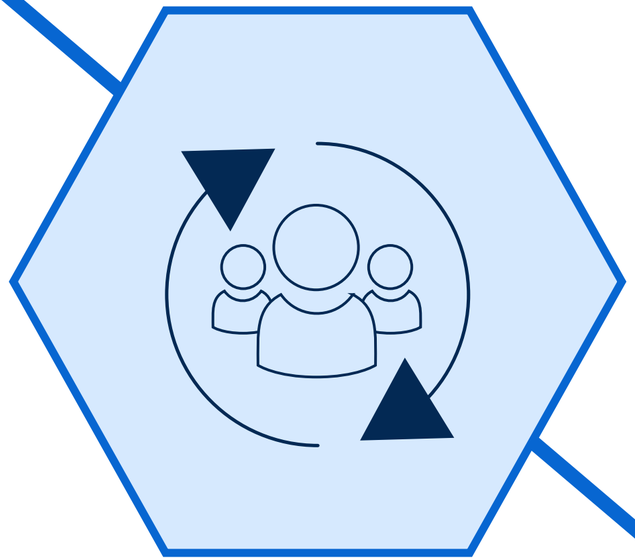
Inclusivity



Ensure accessibility and include diverse perspectives and experiences.

Inclusivity

Design for diverse users and contexts

Inclusivity ensures that predictive models don't disproportionately favor, overlook, or harm certain populations.

Example (Mehta and Giunchilgia (2025))

- A framework and data set for LLMs to better understand expressions unique to Generation Alpha (people born 2010-2025).
- To moderate the safety of online content for a generation who have never lived in a world without strong AI influences.

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Accountability

Proactively identify and mitigate adverse impacts.

Accountability

Define who is responsible for AI actions and outcomes

Accountability is maintained through traceable decisions, timely remediation, and clear lines of ownership for system configuration and for customer impact.

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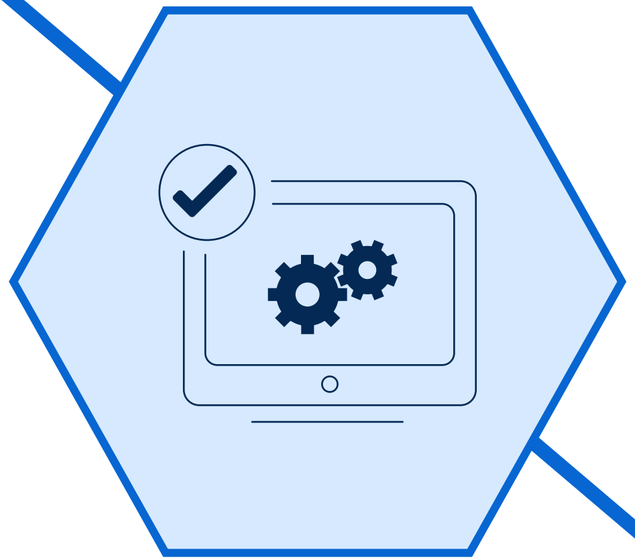
Robustness



Operate reliably and safely while enabling mechanisms that assess and manage potential risks throughout a system's life cycle.

Robustness

Ensure reliability under real-world conditions and in the presence of bad actors

Example:

A campaign is timed for a seasonal launch, but an unexpected viral trend shifts consumer behavior mid-cycle. → AI detects anomalies and recalibrates segments to reflect the real-time sentiment. → The marketing analyst reviews updated logic to confirm fitness with branding goals and customer expectations.

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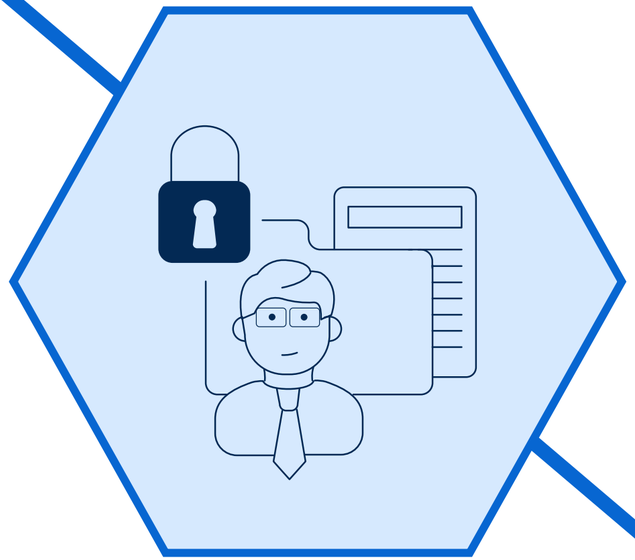
Privacy & Security



Respect the privacy of data subjects and the security of data products.

Privacy and Security

Protect sensitive data and control data access

Example:

AI optimizes email open rates, inferring location history and device usage patterns. Reveals PII.

Analyst has built AI governance into the system, flagging recommendations involving PII

Campaign benefits from smart targeting within ethical boundaries.

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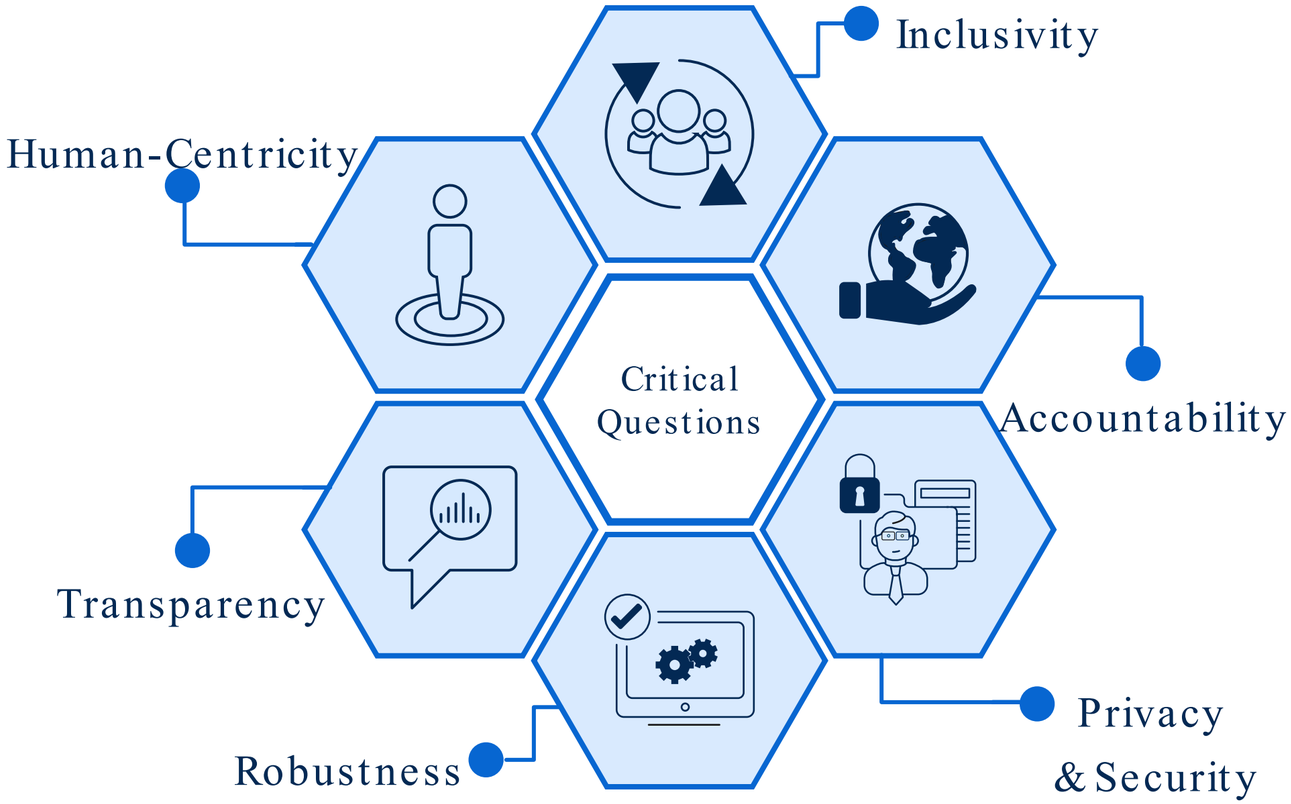
A Sampling of Questions to Ask Your Team

For each principle, describe how AI currently aligns—or does not align—with it.



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Critical Questions

Human-Centricity

- What role do humans play in using, reviewing, or adjusting the model behind this system?
- In what types of business decisions should humans always retain final authority, even when agentic AI makes strong recommendations?
- What risks do you see if agentic systems are allowed to adapt and act on strategies without human review?

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Critical Questions

Transparency

- If a stakeholder saw this model's output, could they understand what information was used when it was produced?
- When predictive models drive campaign decisions, what's the minimum level of transparency that business stakeholders should expect?
- Where does this system offer clarity? Where is it opaque?

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Critical Questions

Inclusivity

- Who benefits most from this model?
- Who was consulted in designing the requirements?
- Is anyone at risk of being disadvantaged?
- What trade-offs might arise between performance optimization and inclusive targeting?

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Critical Questions

Accountability

- Who should be responsible when an Agentic AI tool deploys a biased or harmful campaign message? The developer? The analyst? The business user?
- What processes can help you build to ensure accountability is visible throughout the modeling lifecycle?
- If the system fails or causes harm, what safeguards or ownership structures are in place?

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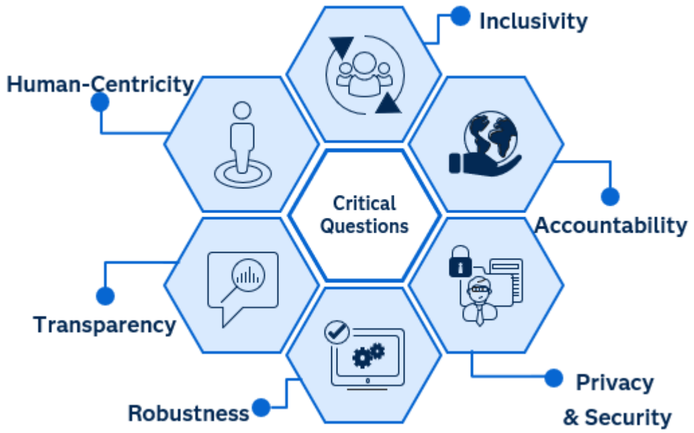
Critical Questions

Robustness

- Can the model respond safely and accurately under unexpected or shifting conditions?
- How might an AI tool trained on historical activity fail during an unexpected event (e.g., major weather event or economic shock)?
- Has this system been tested under different contexts or changing data conditions?
- What backup options exist? Is there a killswitch?



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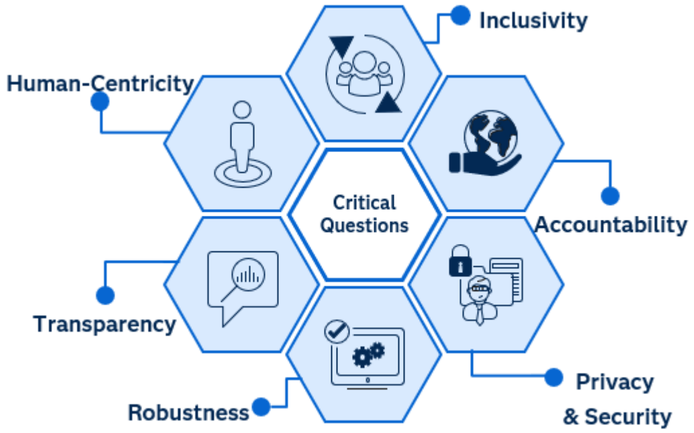
Critical Questions

Privacy & Security

- Are we collecting or using any data that wasn't explicitly consented for this purpose?
- Does the system collect or infer any sensitive data? How is that protected?
- How do you balance personalization with responsible data usage in targeting marketing strategies?



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Ethics Alignment Checklist - A Deeper Dive

Use this checklist during model development and deployment stages to assess alignment with key principles:

Question	Yes/No
Is the model's decision logic fully traceable and documented?	
Have fairness checks been conducted across all key subgroups?	
Are humans consulted or able to intervene in high-impact decisions?	
Are privacy boundaries clear in both data use and predictions?	
Is there a clear record of who built, approved, and deployed the model?	
Is there a record of tests, assumptions, and trade-offs considered by the team?	
Has the model been tested under multiple operational conditions?	
Have stakeholders reviewed potential unintended consequences?	

Tip: If you answer "No" to any, flag that item for improvement before release.

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Probing Questions for Ethical Model Review

Use the critical questions with peers and stakeholders to spark dialogue

Principle	Question
Human Centricity	What decisions should this model make autonomously, and which ones should require human review?
Transparency	If a stakeholder saw this model's output, could they understand what information was used when it was produced?
Inclusivity	Who might be unintentionally excluded or disadvantaged by this model's recommendations?
Accountability	If something goes wrong, who is responsible, and what is the escalation path?
Robustness	Can the model respond safely and accurately under unexpected or shifting conditions?
Privacy and Security	Are we collecting or using any data that wasn't explicitly consented for this purpose?

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2. Ethical Rating

Provide a score (1–5) for each principle where:

1 = major concerns, 5 = excellent alignment. Briefly justify each score.

Principle	Rating (1-5)	Justification
Human Centricity		
Transparency		
Inclusivity		
Accountability		
Robustness		
Privacy and Security		

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3. Improvement Plan

Identify two ethical risks and outline steps to mitigate each risk (e.g., fairness audit, override design, human-in-the-loop, documentation enhancement, etc.).

Risk	Mitigation Strategy

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Model Documentation Guide

Component	Considerations
Model Summary	• Business question and context • Specific use cases and boundaries • High-level flowchart of decision logic
Data Description	• Data sources, formats, and time ranges • Preprocessing steps (for example, imputation, normalization) • Privacy-sensitive fields and consent rationale • Data Lineage and data subject consent
Feature Engineering	• Final feature list with definitions • Feature importance estimates or SHAP values • Notes on fairness audits and bias mitigation

Continued...

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Model Documentation Guide

Component	Considerations
Modeling Technique	• Algorithms or architectures used • Hyperparameter settings and rationale • Trade-offs considered (for example, accuracy vs. fairness)
Evaluation Metrics	• Performance metrics (for example, ROC/AUC, precision, recall, fairness assessment statistics) • Validation methods (cross-validation, holdout) • Domain-specific considerations (for example, uplift for marketing)
Ethical Considerations	• Risks identified during development • Oversight mechanisms and intervention points • Alignment with organizational ethical values

Continued...

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Model Documentation Guide

Component	Considerations
Safeguards	• Monitoring and reporting of disallowed outputs (hate speech, harassment, explicit violence, promotion of self -harm, illegal activities, etc.) • Avoidance of medical, legal, or financial advice without certified expert review • Reporting of queries that aim to bypass security measures or reverse-engineer model parameters • Regular audit and review cadence for flagged incidents and model behavior reports
Deployment and Monitoring	• System integration details • Audit log setup and access controls • Drift detection protocols and retraining schedule • Access Controls or rules
Version Control and Ownership	• Version history and change log • Primary developers, reviewers, and users • Approval dates and review milestones

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Trustworthy AI Features

Manage Data

- ✓ Information Privacy
- ✓ Data Quality
- ✓ Data Lineage
- ✓ Automated Data Exploration
- ✓ Data Suppression
- ✓ Data Masking
- ✓ Synthetic Data Generation

Develop Models

- ✓ Model Interpretability
- ✓ Fairness Assessment & Bias Mitigation
- ✓ Natural Language Insights

Deploy Insights

- ✓ Model Monitoring
- ✓ Model Governance
- ✓ Decision Accountability

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Trustworthy AI Features

- ✓ Information Privacy: Learn whether data contains potentially private information that could be linked to an individual.
- ✓ Data Quality: Identify potential issues that might have unintended consequences during modeling.
- ✓ Data Lineage: View data assets in context of their sources, outputs, and the relationships among them.
- ✓ Automated Data Exploration: Profile data automatically to find the most important inputs, identify suspicious variables and assess the distribution of features in the training data set.
- ✓ Data Suppression: Suppress values in a data set that could be used to infer sensitive information.
- ✓ Data Masking: Anonymize and mask data in selected columns.
- ✓ Synthetic Data Generation: Generate tabular synthetic data using generative adversarial networks(GANs) and synthetic minority oversampling technique (SMOTE).

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Trustworthy AI Features

Manage Data

- ✓ Information Privacy
- ✓ Data Quality
- ✓ Data Lineage
- ✓ Automated Data Exploration
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Develop Models

- ✓ Model Interpretability
- ✓ Fairness Assessment & Bias Mitigation
- ✓ Natural Language Insights

Deploy Insights

- ✓ Model Monitoring
- ✓ Model Governance
- ✓ Decision Accountability

- ✓ Model Interpretability: Understand the results of AI models with techniques like PD, ICE, LIME, and HyperSHAP.
- ✓ Fairness Assessment & Bias Mitigation: Uncover potential differences in model performance and predictions for different groups, and proactively reduce biases by training models that are fair across sensitive variables.
- ✓ Natural Language Insights: Automatically generate explanations of data and models written in simple business language understood by any audience.

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Trustworthy AI Features

Manage Data

- ✓ Information Privacy
- ✓ Data Quality
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Develop Models

- ✓ Model Interpretability
- ✓ Fairness Assessment & Bias Mitigation
- ✓ Natural Language

Deploy Insights

- ✓ Model Monitoring
- ✓ Model Governance
- ✓ Decision Accountability

- ✓ Model Monitoring: Automatically monitor deployed models to ensure that they remain accurate, fair, and relevant over time.
- ✓ Model Governance: Govern all analytical assets in a centralized, searchable repository that ensures transparency, traceability, and complete visibility into analytical processes.
- ✓ Decision Accountability: Create, approve, annotate, deploy, and audit decisioning processes in a single visual interface.

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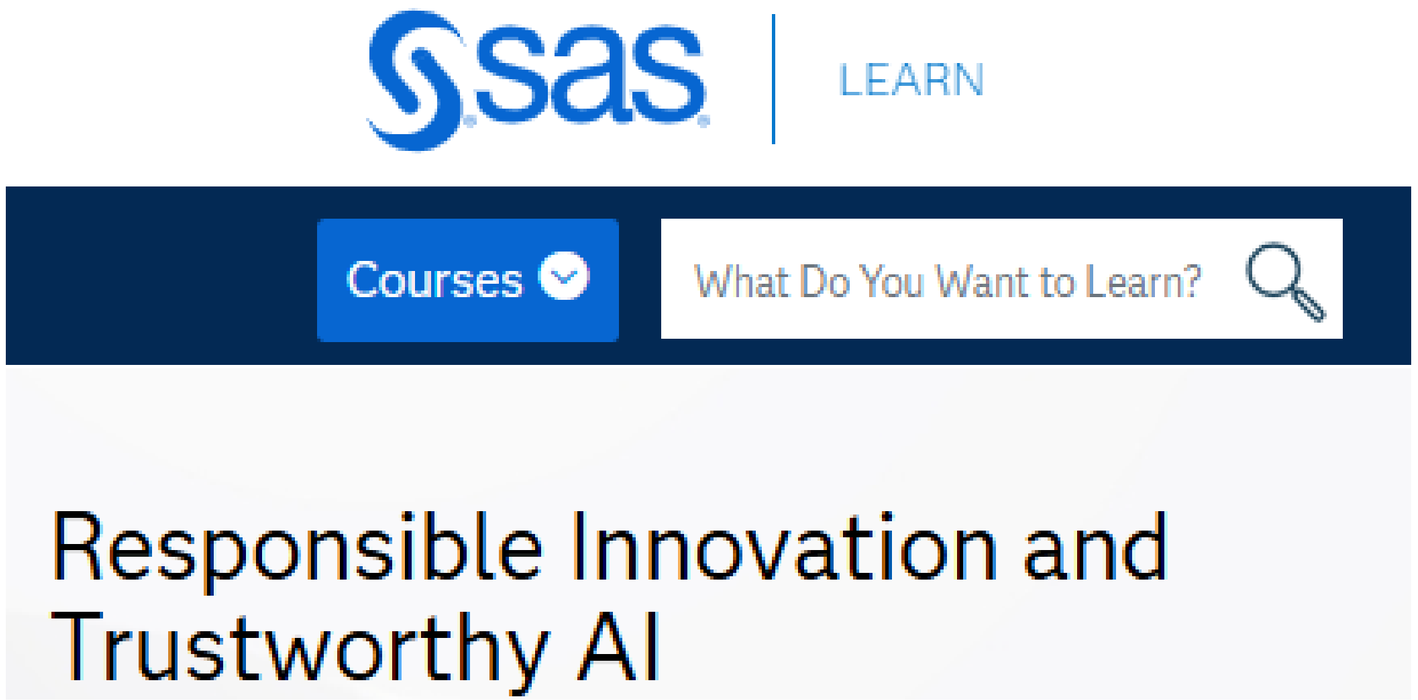
To Learn More

https://learn.sas.com/course/view.php?id=6443



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